Syllabus – Module 5: Remote sensing, Weather forecasting and Navigation Satellites

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Topic 01: Development of Satellite Navigation Systems

Various navigation methods have been used over the ages including marking of trails using stones and twigs, making maps, making use of celestial bodies (sun, moon and stars), monumental landmarks and using instruments like the magnetic compass, sextant, etc. These traditional methods were superceded by ground-based radio navigation techniques in the early 20th century. These ground-based systems were widely used during World War II. These systems, however, could only provide accurate positioning services in small coverage areas. Accuracy reduced with an increase in the coverage area. Satellite-based navigation systems were developed to provide accurate as well as global navigation services simultaneously. These systems were originally developed for military operations, but their use for civilian applications soon became common place. Initially, the systems developed were based on the 'Doppler effect'. Later 'trileration'- based systems came into the picture.

Doppler Effect based Satellite Navigation Systems:

The first satellite navigation system was the Transit system developed by the US Navy and John Hopkins University of the USA back in the early 1960s. The first satellite in the system, Transit I, was launched on 13 April 1960. It was also the first satellite to be launched for navigation applications. The system was available for military use in the year 1964 and to civilians three years later in 1967. The system employed six satellites (three active satellites and three in-orbit spares) in circular polar LEOs at altitudes of approximately 1000 km. The last Transit satellite was launched in the year 1988. The main limitations of the system were that it provided only two-dimensional services and was available to users for only brief time periods due to low satellite altitudes. Moreover, high speed receivers were not able to use the system. The system was terminated in the year 1996. The Transit system was followed by the Nova navigation system, which was an improved system having better accuracy. Russians launched their first navigation satellite, Kosmos-158, in the year 1967, seven years after the first American navigation satellite launch. The satellite formed a part of the Tsyklon system. It provided services similar to that provided by the Transit system. It was operational until 1978. The system was superceded by the Parus and the Tsikada systems. Parus is a military system comprising satellites in six orbital planes spaced at 30◦ longitude intervals, thus having an angular coverage of 180◦. Ninety-eight satellites have been launched in the Parus system with the last satellite Parus 98 launched on 21 July 2009. The system is operational and is mainly used for data relay and store-dump communication applications. Tsikada is a civilian system covering the rest of the 180◦. It comprised satellites in four orbital planes at 45◦ intervals. Twenty Tsikada satellites have been launched with the last satellite launched on 21 January 1995.

All the systems discussed above are based on the 'Doppler effect'. The satellite transmits microwave signals containing information on its path and timing. The pattern of the Doppler shift of this signal transmitted by the satellite is measured as it passes over the receiver (Figure 12.1). The Doppler pattern coupled with the information on the satellite orbit and timing establishes the location of the receiver station precisely. One satellite signal is sufficient for determining the receiver locations. However, the systems mainly transmit two frequencies as it improves the accuracy of the system. As an example, the Transit system transmitted signals at 150 MHz and 400 MHz frequencies. The positioning accuracy was around 500 m for single frequency users and 25 m for dual frequency users.

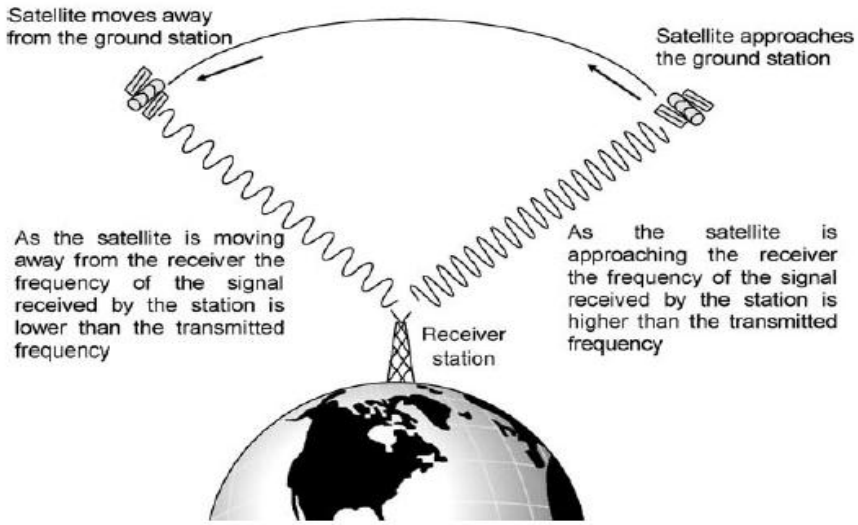

Figure 12.1 Principle of operation of Doppler effect based satellite navigation systems

Trilateration-based Satellite Navigation Systems:

Doppler-based navigation systems have given way to systems based on the principle of 'trileration', as they offer global coverage and have better accuracy as compared to the Doppler-based systems. In this case, the user receiver's position is determined by calculating its distance from three (or four) satellites whose orbital and the timing parameters are known. The receiver is at the intersection of the invisible spheres, with the radius of each sphere equal to the distance between a particular satellite and the receiver, with the centre being the position of that satellite (Figure 12.2). Two such systems are in operation today, namely the Global Positioning System (GPS) of the USA and the Global Navigation Satellite System (GLONASS) of Russia. The first test satellite of the constellation was launched on 28 December 2005, while the second test satellite was launched on 26 April 2008. The third test satellite will be launched in the near future. These satellites will characterize the critical technologies of the system.

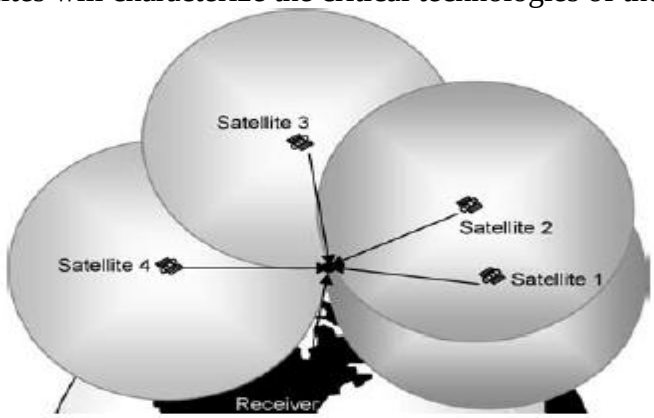

Figure 12.2 Principle of operation of trileration-based satellite navigation systems

1.1 Development of the Global Positioning System (GPS)

The first effort in this area began in the year 1972, with the launch of Timation satellites. These satellites provided time and frequency transfer services. Three satellites were launched in this series. The third satellite acted as a technology demonstrator for the Global Positioning System (GPS) Program, also known as the Navigation Satellite Timing and Ranging (NAVSTAR) Program. The GPS was the first operational navigation system that provided continuous positioning and timing information anywhere in the world. It was developed by the US Department of Defense (DoD) for use in military operations. It is now a dual-use system, used for both military as well as civilian applications. The GPS receivers calculate their location on the basis of ranging, timing and position information transmitted by GPS satellites [the GPS satellites\ transmit information at two frequencies, 1575.42 MHz (L1) and 1227.6 MHz (L2)].

The first GPS satellite was launched on 22 February 1978. It marked the beginning of first generation GPS satellites, referred to as Block-I satellites (Figure 12.3). Eleven satellites were launched in this block and were mainly used for experimental purposes. These satellites were out of service by the year 1995. The second generation of GPS satellites (Figure 12.4) comprised Block-II and Block-IIA satellites. Block-IIA satellites were advanced versions of Block-II satellites. A total of 28 Block-II and Block-IIA satellites (nine satellites in the Block-II series and 19 satellites in the Block-IIA series) were launched over the span of eight years, from 1989 to 1997. The GPS system was declared fully functional on 17 July 1995, ensuring the availability of at least 24 operational, non-experimental GPS satellites. Currently, third generation GPS satellites, referred to as Block-IIR (Figure 12.5) satellites, are being launched. The first satellite in this series was launched in the year 1997, with 21 satellites planned to be launched in this block. Till December 2009, 13 Block-IIR satellites had been launched. One of the potential advantages of Block-IIR satellites over the Block-II and -IIA satellites is that they have reprogrammable satellite processors enabling upgradation of satellites while in orbit. These satellites can calculate their own positions using intersatellite ranging techniques. Moreover, they have more stable and accurate clocks on board as compared to the Block-II and Block-IIA satellites. Block-IIR satellites have three Rubidium atomic clocks (having an accuracy of 1 second in 300 000 years), whereas Block-II and Block-IIA satellites have two Cesium atomic clocks (having an accuracy of 1 second in 160 000 years) and two Rubidium atomic clocks (having an accuracy of 1 second in 300 000 years). Eight of the planned Block-IIR satellites have been improved further and are renamed Block-IIR-M satellites. These satellites will carry a new military code on both the frequencies (L1 and L2) and a new civilian code on the L2 frequency. The dual codes will provide increased resistance to jamming and the new civilian code will provide better accuracy to civilian users by increasing capability to compensate for atmospheric delays. Seven Block-IIR-M satellites have been launched till date.

Block-IIR-M satellites will be followed by Block-IIF satellites, with 12 Block-IIF satellites (Figure 12.6) planned to be launched by the year 2011. These satellites will have a third carrier signal, L5, at 1176.45 MHz. They will also have a larger design life, fast processors with more memory and a new civilian code. The GPS-III phase of satellites are in the planning stage. These satellites will employ spot beams, enabling the system to have better position accuracy (less than a meter). They will be positioned in three orbital planes having non-recurring orbits.

Figure 12.3 Block-I GPS satellite (Courtesy: NASA)

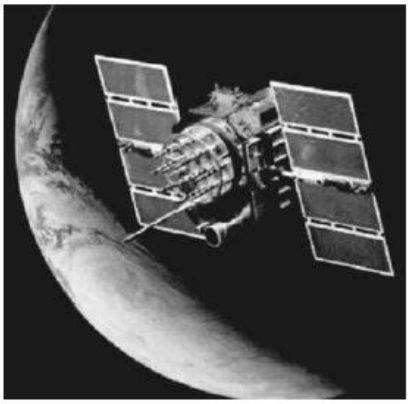

Figure 12.4 Block-II and -IIA GPS satellites (Courtesy: NASA)

Figure 12.5 Block-IIR GPS satellites (Reproduced by permission of Lockheed Martin)

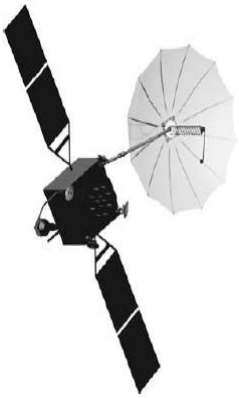

Figure 12.6 Block-IIF GPS satellites (Reproduced by permission of © Aerospace Corporation)

Topic. 2: Global Positioning System (GPS)

The GPS comprises of three segments, namely the space segment, control segment and user segment. All the three segments work in an integrated manner to ensure proper functioning of the system.

Space Segment:

The space segment comprises of a 28 satellite constellation out of which 24 satellites are active satellites and the remaining four satellites are used as in-orbit spares. The satellites are placed in six orbital planes, with four satellites in each plane. The satellites orbit in circular medium Earth orbits (MEO) at an altitude of 20 200 km, inclined at 55◦ to the equator (Figure 12.10). The orbital period of each satellite is around 12 hours (11 hours, 58 mins). The MEO orbit was chosen as a compromise between the LEO and GEO orbits. If the satellites are placed in LEO orbits, then a large number of satellites would be needed to obtain adequate coverage. Placing them in GEO orbits would reduce the required number of satellites, but will not provide good polar coverage. The present constellation makes it possible for four to ten satellites to be visible to all receivers anywhere in the world and hence ensure worldwide coverage.

All GPS satellites are equipped with atomic clocks having a very high accuracy of the order of a few nanoseconds (3 ns in a second). These satellites transmit signals, synchronized with each other on two microwave frequencies of 1575.42 MHz (L1) and 1227.60 MHz (L2). These signals provide navigation and timing information to all users worldwide. The satellites also carry nuclear blast detectors as a secondary mission, replacing the 'Vela' nuclear blast surveillance satellites. The satellites are powered by solar energy. They have back-up batteries on board to keep them running in the event of a solar eclipse. The satellites are kept in the correct path with the help of small rocket boosters, a process known as 'station keeping'.

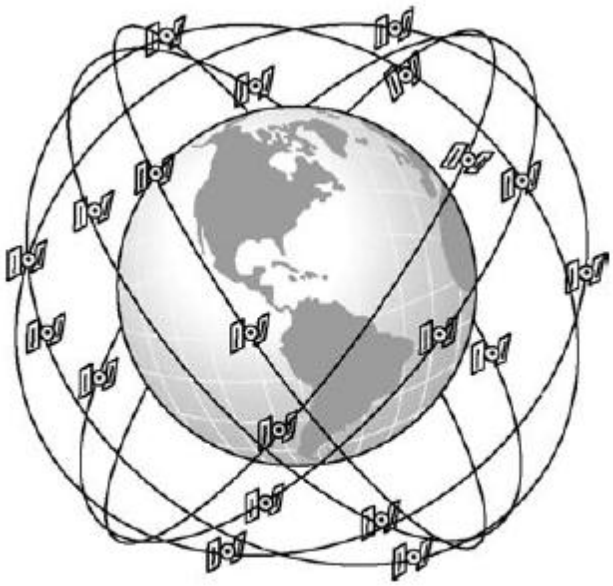

Figure 12.10 Space segment of GPS

Control Segment:

The control segment of the GPS system comprises a world wide network of five monitor stations, four ground antenna stations and a master control station. The monitor stations are located at Hawaii and Kwajalein in the Pacific Ocean, Diego Garcia in the Indian Ocean, Ascension Island in the Pacific Ocean and Colorado Springs, Colorado. There is a master control station (MCS) at Schriever Air Force Base in Colorado that controls the overall GPS network. The ground antenna stations are located at Diego Garcia in the Indian Ocean, Kwajalein in the Pacific Ocean, Ascension Island in the Pacific Ocean and at Cape Canaveral, USA. Figure 12.11 shows the locations of the stations of the control segment. Each of the monitor stations is provided with high fidelity GPS receivers and a Cesium oscillator to continuously track all GPS satellites in view. Data from these stations is sent to the MCS which computes precise and updated information on satellite orbits and clock status every 15 minutes. This tracking information is uploaded to GPS satellites through ground antenna stations once or twice per day for each satellite using S band signals. This helps to maintain the accuracy and proper functioning of the whole system. The ground antenna stations are also used to transmit commands to satellites and to receive satellite telemetry data. Figure 12.12 describes the functioning of the control segment.

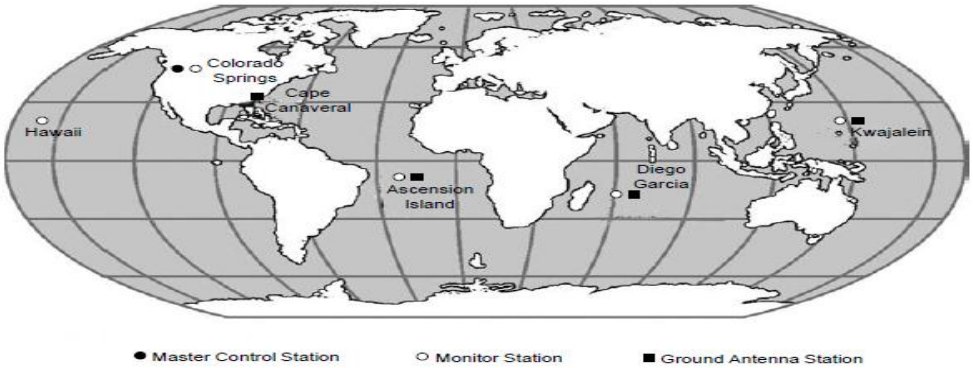

Figure 12.11 Control segment of GPS

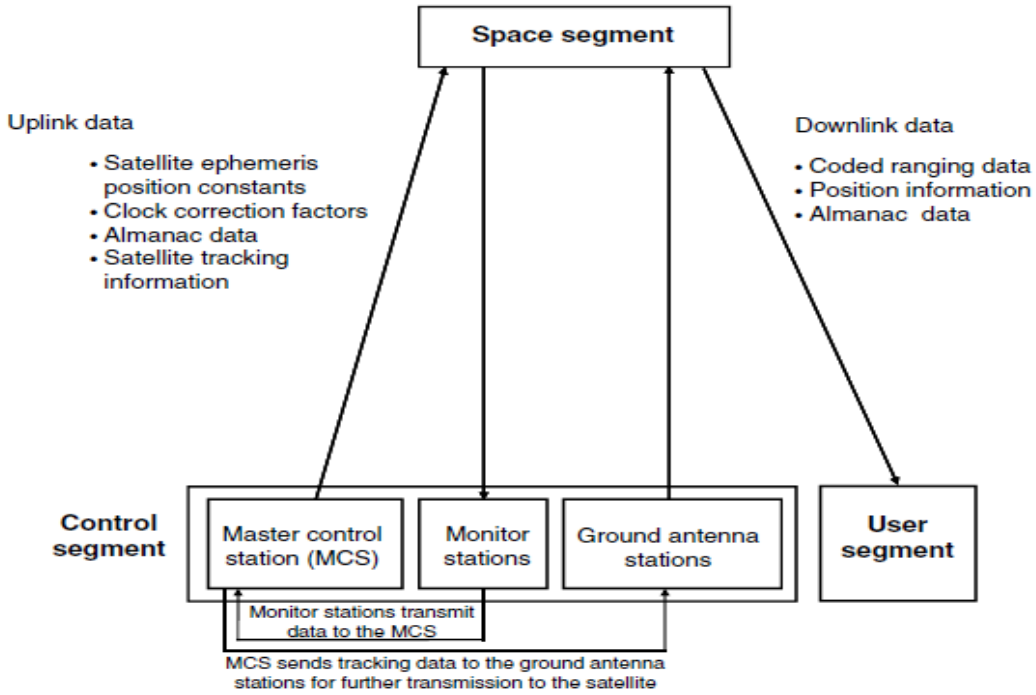

Figure 12.12 Operation of the control segment of GPS system

User Segment:

The user segment includes all military and civil GPS receivers intended to provide position, velocity and time information. These receivers are either hand-held receivers or installed on aircraft, ships, tanks, submarines, cars and trucks. The basic function of these receivers is to detect, decode and process the GPS satellite signals. Most receivers trace the path of the user as they move. Certain advanced receivers also tell the user the distance they have travelled, their speed and time of travel. They also tell the estimated time of arrival at the current speed when fed with destination coordinates. Moreover, there is no limit to the number of users using the

system simultaneously. Figure 12.13 shows the photograph of a commonly used GPS receiver.

Figure 12.13 GPS receiver (Reproduced by permission of © Randy Bynum/www.nr6ca.org)

GPS receivers comprise three functional blocks:

1. **Radio frequency front end:** The front end comprises one or more antennas to receive the GPS signal, filters and amplifiers to discriminate the wanted signal from noise and a downconverter to remove the carrier signal.
2. **Digital signal processing block**: It correlates the signals from satellites with signals stored in the receiver to identify the specific GPS satellite and to calculate pseudo-ranges.
3**. Computing unit**: This unit determines position, velocity and other data. The display format is also handled by the computing unit.

Figure 12.14 explains the functionality of the GPS system. GPS satellites transmit coded information that is used for range calculations. Range and satellite position information from three or four satellites is used to calculate the position of the receiver.

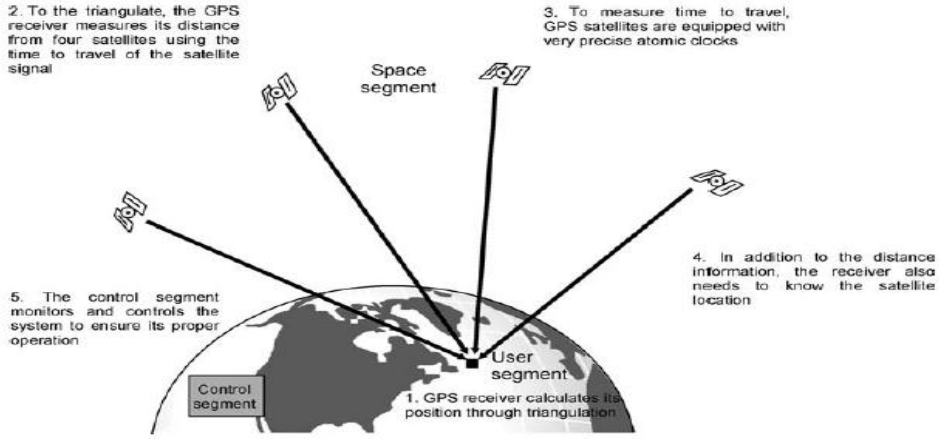

Figure 12.14 Operation of the GPS navigation system

2.1: Working Principle of the GPS

Principle of Operation:

The basic principle of operation of the GPS is that the location of any point can be determined if its distance is known from four objects or points with known positions. Theoretically, if the distance of a point is known from one object, then it lies anywhere on a sphere with the object as the centre having a radius equal to the distance between the point and the object [Figure 12.15

(a)]. If the distance of the point is known from two objects, then it lies on the circle formed by the intersection of two such spheres [Figure 12.15 (b)]. The distance from the third object helps in knowing that the point is located at any of the two positions where the three spheres intersect [Figure 12.15 (c)]. The information from the fourth object reveals the exact position where it is located i.e., at the point where the four spheres intersect. In the GPS, the position of any receiver is determined by calculating its distance from four satellites. This distance is referred to as the 'Pseudorange'. The information from three satellites is sufficient for calculating the longitude and the latitude positions; however, information from the fourth satellite is necessary for altitude calculations. Hence, if the receiver is located on Earth, then its position can be determined on the basis of information of its distance from three satellites. The GPS is also a source of accurate time, time interval and frequency information anywhere in the world with unprecedented precision. The GPS uses a system of coordinates called WGS-84, which stands for World Geodetic System 1984. It produces maps having a common reference frame for latitude and longitude lines. The system uses time reference from the US Naval Observatory in Washington DC in order to synchronize all timing elements of the system.

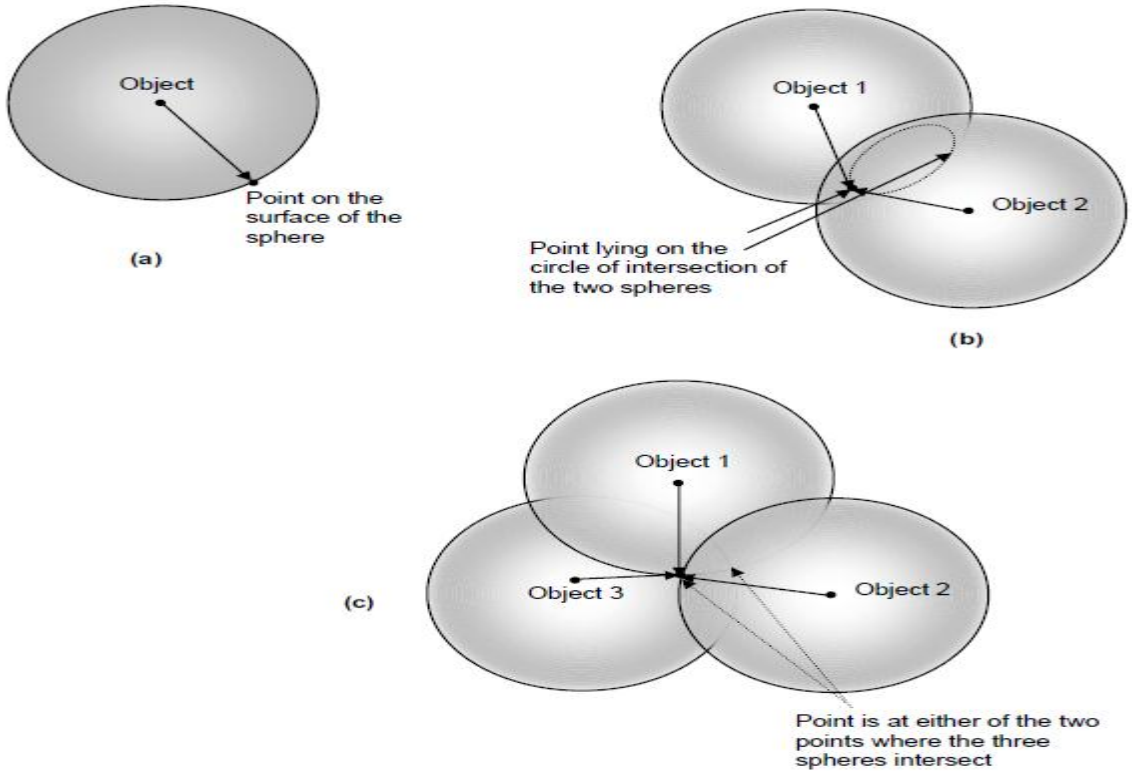

Figure 12.15 Determination of the position of any point

2.2: Applications of Satellite Navigation Systems

Satellite navigation represents one of the dual use space technologies that has found extensive applications both in military and civilian fields. Some of the main military application areas include weapon guidance, navigation, tracking, etc. Civilian applications include construction and surveying, seismic surveying, airborne mapping, vehicle navigation, automotive, marine, military and aviation surveying. They are also used in endeavours like aerial refuelling, rendezvous operations, geodetic surveying and various search and rescue operations.

Military Applications:

Satellite navigation systems have proved to be a valuable aid for military forces. Military forces around the world use these systems for diverse applications including navigation, targeting, rescue, disaster relief, guidance and facility management, both during wartime as well as peacetime. GPS and GLONASS receivers are used by soldiers and also have been incorporated on aircraft, ground vehicles, ships and spacecraft.

1. Navigation. Navigation systems are invaluable for soldiers to navigate their way in unfamiliar enemy territory (Figure 12.28). They can also be used by special forces and crack teams to reach and destroy vital enemy installations. As an example, GPS receivers were used extensively by the US soldiers during Operation Desert Storm and Operation Iraqi Freedom.
2. Tracking. The services of navigation satellites are also utilized to track potential targets before they are declared hostile to be engaged by various weapon platforms. The tracking data is fed as input to modern weapon systems such as missiles and smart bombs, etc.
3. Bomb and missile guidance. The GPS and GLONASS systems are used to guide bomb and missiles to targets and position artillery for precise fire even in adverse weather conditions. Cruise missiles commonly used by the USA use multichannel GPS receivers to determine accurately their location constantly while in flight. The multiple launched rocket system (MLRS) vehicle uses GPS-based inertial guidance to position itself and aim the launch box at a target in a very short time (Figure 12.29).
4. Rescue operations. Satellite navigation systems prove invaluable to the military for determining the location of causality during operations and in navigating rescue teams to the site.
5. Map updation. These systems augment the collection of precise data necessary for quick and accurate map updation.

Figure 12.28 Soldiers using GPS receivers (Reproduced by permission of © The Aerospace Corporation)

Figure 12.29 GPS-based inertial guidance system (Reproduced by permission of © The Aerospace Corporation)

Civilian Applications:

Initially developed for military applications, satellite navigation systems soon became commonplace for civilian applications as well. In fact, civil applications outnumber military uses in terms of range of applications, number of users and total market value. Satellite navigation systems are finding newer and newer commercial applications due to decreasing cost, size and

introduction of new features. Civilian applications include marine and aviation navigation, precision timekeeping, surveying, fleet management, mapping, construction & surveying, aircraft approach assistance, geographic information system (GIS), vehicle tracking, natural resource and wildlife management, disaster management and precision agriculture etc.

1. For mapping and construction. Mapping, construction and surveying companies use satellite navigation systems extensively as they can provide real time submetre and centimeter level positioning accuracy in a cost-effective manner. They are mainly used in road construction, Earth moving and fleet management applications. For these applications, receivers along with wireless communication links and computer systems are installed on board the Earth moving machines (Figure 12.30). The required surface information is fed to this machine. With the help of real time position information, an operator obtains information as to whether the work is in accordance with the design plan or not. Satellite navigation systems are also used for telecom power placement, laying of pipelines, flood plane mapping, oil, gas and mineral exploration and in glacier monitoring.

2. Saving lives and property. Many police, fire and emergency medical service units employ GPS receivers to determine which available police car, fire truck or ambulance is nearest to the emergency site, enabling a quick response in these critical situations. GPS-equipped aircraft monitor the location of forest fires exactly, enabling the fire supervisors to send firefighters to the required spot on time.

3. Vehicle tracking and navigation. Vehicle tracking is one of the fastest growing satellite navigation applications today. Many fleet vehicles, public transportation systems, delivery trucks and courier services use GPS and GLONASS receivers to monitor their locations at all times. These systems combined with digital maps are being used for vehicle navigation applications. These digital maps contain information like street names and directions, business listings, airports and other important landmarks. Such units provide useful information about the car's position and the best travel routes to a given destination by linking itself to a built-in digital map (Figure 12.31).

One of the emerging uses of the GPS and GLONASS systems is for air traffic control (ATC). Here these systems are used for navigating and tracking the aircraft while in flight (Figure 12.32). This helps in efficient routing (and hence in saving fuel) and in closer spacing of plane routes in the air. They are also used in maritime navigation applications for vehicle tracking and traffic management, etc.

4. Environmental monitoring. GPS-equipped balloons monitor holes in the ozone layer across the globe. Buoys tracking major oil spills transmit data using the GPS to guide the clean-up operations. GPS systems are also used in wildlife management and insect infestation. They are also used for determination of forest boundaries.

5. Monitoring structural deformations. Navigation systems are used for measuring deformations on the Earth's crust. This helps in the prediction of earthquakes and volcanic eruptions. Geophysicists have been exploiting the GPS since the mid-1980s to measure continental drift and the movement of the Earth's surface in geologically active regions. They are also used for monitoring the deformation of dams, bridges and TV towers.

Figure 12.30 Use of GPS satellites for mapping and construction (Reproduced by permission of © Leica Geosystems)

Figure 12.31 Satellite navigation systems for cars (Reproduced by permission of Paul Vlaar)

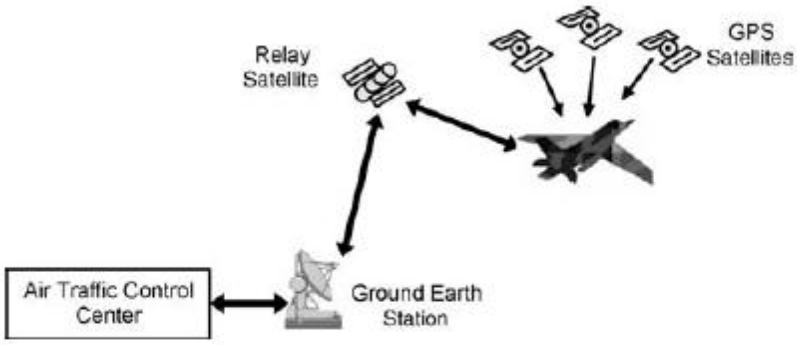

Figure 12.32 Use of navigation satellites in air traffic control

***6. Archeology*.** Archeologists, biologists and explorers are using the satellite navigation systems to locate ancient ruins, migrating animal herds and endangered species.

7. Utility industry. Navigation systems are of tremendous help to the utility industry companies like electric, gas, water companies, etc. Up-to-date maps provided by the navigation systems help these companies to plan, build and maintain their assets.

8. Precision farming. Farming systems employ navigation receivers to provide precise guidance for field operations and in the collection of map data on tillage, planting, weeds, insect and disease infestations, cultivation and irrigation.

9. Precise timing information. GPS satellites provide precise and accurate timing information within 100 ns of the universal time coordinated (UTC) atomic clock. The receiver required for this application is different as well as more expensive than the standard GPS receivers. It is used in applications such as telecommunications and scientific research. It is also used for precise transfer of time between the world's timing centres and helps in tracking deep space vehicles. It is a source of precise time for various military and intelligence operations.

Question Bank

Module 05: Navigation Satellites

1. What are the military and civilian applications of a satellite navigation system. (5M)
2. Explain Global positioning system along with segments and working principle. (15M)